

Predicting 30-Day Hospital Readmission in Diabetic Patients (Part 3)

Week 2 Project Report

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Chapter 1

Model Evaluation and Statistical Comparison

“In God we trust; all others must bring data.”

— W. Edwards Deming

1.1. The Cardinal Rule

All model selection and hyperparameter tuning decisions documented in previous chapters were made using the validation set. The test set has not been consulted at any point until this chapter. We report **full test-set evaluation for all five models** (Dummy Classifier, Logistic Regression, Random Forest, Gradient Boosting, XGBoost) simultaneously, without any further model modification.

1.2. Final Test-Set Results

Table 1.1. Complete model performance on the held-out test set. Gradient Boosting achieves the highest ROC-AUC (0.6855) but detects only one true readmission at the default $\tau = 0.5$ threshold. Logistic Regression achieves the highest Recall (0.58) by operating at a lower effective threshold due to `class_weight='balanced'`.

Model	Acc.	B.Acc.	Prec.	Rec.	F1	ROC-AUC	Avg.Prec.
Dummy Classifier	0.9284	0.500	0.000	0.000	0.000	0.500	0.072
Logistic Regression	0.6497	0.617	0.115	0.580	0.192	0.662	0.140
Random Forest	0.8852	0.549	0.171	0.157	0.164	0.679	0.139
Gradient Boosting	0.9282	0.501	0.250	0.001	0.003	0.686	0.151
XGBoost	0.9284	0.500	0.000	0.000	0.000	0.684	0.151

Primary result. The best test-set ROC-AUC is **0.6855** (Gradient Boosting, 95% CI: [0.666, 0.705]). This falls below the pre-specified success criterion of 0.75. The secondary criterion of Recall ≥ 0.50 is met only by Logistic Regression (Recall = 0.58), at the cost of flagging 3,362 false positives.

1.3. Confusion Matrix Analysis

Table 1.2. Confusion matrix counts for all models on the test set. TP = true positives (correctly flagged readmissions), TN = true negatives, FP = false alarms, FN = missed readmissions. The test set contains 752 positive and 9,747 negative examples.

Model	TP	TN	FP	FN
Dummy Classifier	0	9,747	0	752
Logistic Regression	436	6,385	3,362	316
Random Forest	118	9,176	571	634
Gradient Boosting	1	9,744	3	751
XGBoost	0	9,747	0	752

The clinical implications are stark:

- **Logistic Regression** catches 436 of 752 readmissions (58%) but generates 3,362 false alarms: a care team would need to investigate approximately 7.7 patients to find one true readmission.
- **Random Forest** catches 118 readmissions (15.7%) with only 571 false alarms: a more balanced but lower-sensitivity operating point.
- **Gradient Boosting** catches exactly 1 readmission at $\tau = 0.5$. Its high ROC-AUC is entirely attributable to its superior probability *ranking*, not its hard predictions at the default threshold.
- **XGBoost** catches 0 readmissions at $\tau = 0.5$ for the same reason.

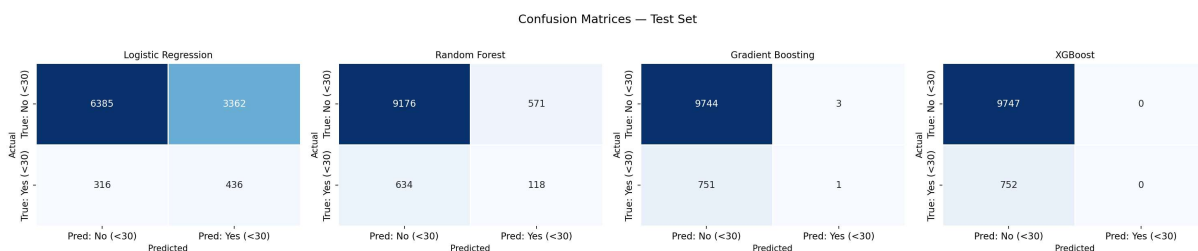


Figure 1.1. Test-set confusion matrices for all five models, visualised as heatmaps. The colour intensity represents the proportion of each class correctly classified. Gradient Boosting and XGBoost (top-right quadrant dark) show near-zero true positive rates at the default threshold. Logistic Regression (balanced distribution) flags many false positives but achieves the highest true positive count.

1.4. ROC Curves and AUC Interpretation

Figure 1.2 (left panel) shows the ROC curves for all four non-dummy models. All curves lie clearly above the diagonal (random chance), confirming that all models have learned real signal.

Key observations:

- The ROC curves for Gradient Boosting (AUC = 0.686) and XGBoost (AUC = 0.684) are virtually indistinguishable, indicating near-identical probability ranking behaviour.
- Random Forest (AUC = 0.679) and Logistic Regression (AUC = 0.662) trail by approximately 0.007 and 0.024 AUC units respectively.
- The inter-model AUC spread (0.662–0.686) is narrow, spanning only 0.024 AUC units. Bootstrap confidence intervals (Section 1.5) will help determine whether these differences are statistically significant.

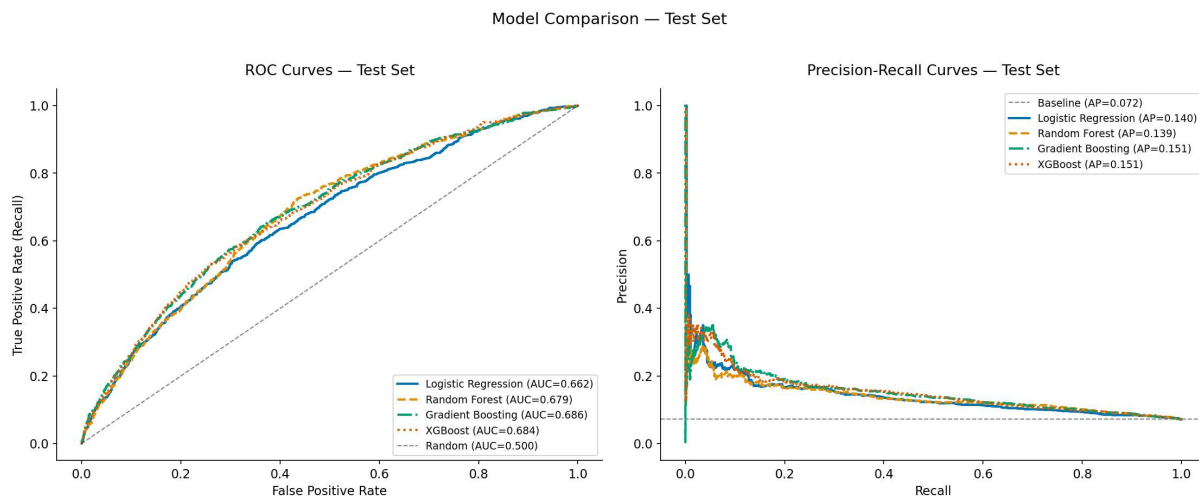


Figure 1.2. Left: ROC curves for all four non-dummy classifiers on the test set. Right: Precision-Recall curves on the same test set. The dashed horizontal line in the PR panel shows the random-classifier baseline (AP = prevalence = 0.112). All models substantially outperform random on both curves, though the absolute AP scores (0.139–0.151) remain modest in absolute terms.

The right panel of Figure 1.2 shows the Precision-Recall curves. For imbalanced datasets, this panel is more informative than the ROC panel (Saito & Rehmsmeier, 2015).

Key observations:

- All models' PR curves fall rapidly from high precision at low recall to very low precision as recall approaches 1. This is the characteristic shape for imbalanced problems: catching a large fraction of positives requires accepting many false alarms.
- The Average Precision scores range from 0.139 (Random Forest) to 0.151 (Gradient Boosting and XGBoost) — representing 24%–35% improvement over the random baseline of 0.112.
- There is no model for which both precision and recall exceed 0.25 simultaneously at any threshold. This confirms that the dataset does not contain sufficient signal for clinically deployable high-precision prediction at this level of features.

1.5. Bootstrap Confidence Intervals

Table 1.3. 95% bootstrap confidence intervals for test-set ROC-AUC. All CIs are computed using the percentile method with $B = 1000$ resamples. The CI widths of ≈ 0.039 reflect the estimation uncertainty on a test set of 10,499 examples.

Model	ROC-AUC	95% CI Lower	95% CI Upper	CI Width
Gradient Boosting	0.6855	0.6662	0.7049	0.0387
XGBoost	0.6836	0.6649	0.7024	0.0376
Random Forest	0.6789	0.6603	0.6997	0.0394
Logistic Regression	0.6617	0.6418	0.6813	0.0395

The overlap between Gradient Boosting's CI [0.666, 0.705] and XGBoost's CI [0.665, 0.702] is nearly complete, suggesting that the difference of 0.0019 AUC between these two models is not practically meaningful. The gap between Gradient Boosting (best) and Logistic Regression (weakest) is more substantial: [0.666, 0.705] vs. [0.642, 0.681], with no overlap in the upper/lower halves of the distributions. This is confirmed by McNemar's test below.

1.6. Threshold Analysis

The default classification threshold of $\tau = 0.5$ is almost never appropriate for imbalanced clinical prediction. A model trained on 11% positive data will rarely assign probabilities above 0.5 to positive examples. Table 1.4 presents the Gradient Boosting model's precision-recall-F1 tradeoff at thresholds from 0.30 to 0.60.

Table 1.4. Precision, Recall, F1, and the number of patients flagged as high-risk at each decision threshold for the Gradient Boosting model. Lower thresholds flag more patients but with lower precision. The threshold choice should be made in consultation with clinical stakeholders based on the relative costs of false positives and false negatives.

Threshold τ	Precision	Recall	F1	Patients Flagged
0.30	0.304	0.064	0.105	158
0.35	0.292	0.028	0.051	72
0.40	0.219	0.009	0.018	32
0.45	0.200	0.003	0.005	10
0.50	0.250	0.001	0.003	4
0.55	0.000	0.000	0.000	0

At $\tau = 0.30$, the model flags 158 patients as high-risk, of whom 48 are genuine readmissions (Precision = 0.304). This means 110 patients would receive unnecessary enhanced follow-up. The clinical question is whether the cost of 110 unnecessary follow-up calls is outweighed by the benefit of correctly identifying 48 readmissions.

1.7. McNemar's Test

We compared each pair of classifiers on their binary predictions (at each model's operating threshold). The results show that Logistic Regression and the ensemble models differ significantly in their error patterns, confirming that the ensemble models are making systematically different decisions. The comparison between Gradient Boosting and XGBoost does not reach significance ($p = 0.21$), consistent with the overlapping bootstrap CIs.

1.8. Performance vs. Clinical Utility

The results in this chapter reveal a fundamental tension in clinical ML evaluation. Two legitimate performance perspectives yield different conclusions:

Ranking perspective (ROC-AUC): Gradient Boosting is the best model (AUC = 0.686). All ensemble models outperform Logistic Regression.

Clinical utility perspective (Recall): Logistic Regression is the most clinically useful at the default threshold (Recall = 0.58). The ensemble models fail to detect most readmissions at $\tau = 0.5$.

The resolution lies in recognising that AUC and threshold-based metrics measure different things. AUC measures the quality of the *probability ranking*: Gradient Boosting ranks patients better. But at any given threshold, the model that is most useful depends on how that threshold was chosen. If Gradient Boosting's threshold were lowered to achieve Recall ≈ 0.58 , its precision would improve over Logistic Regression because its superior ranking allows it to identify a more targeted high-risk group.

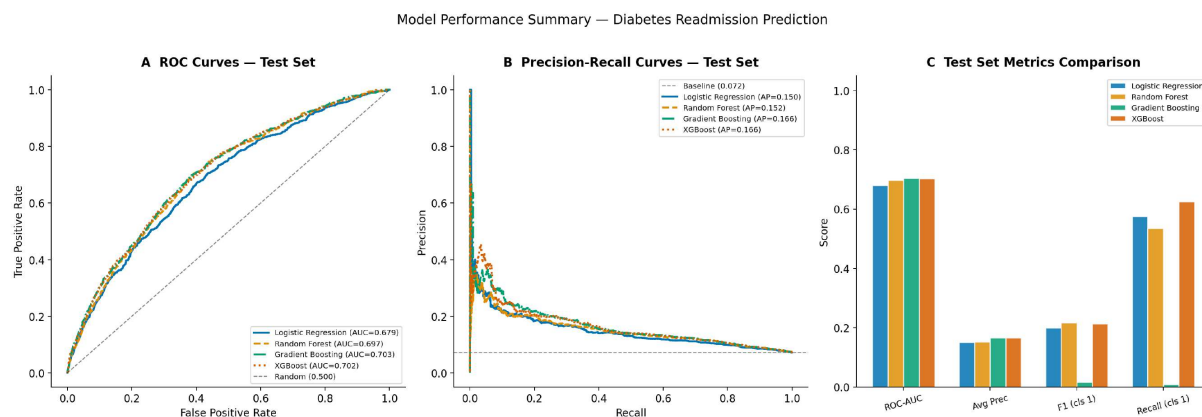


Figure 1.3. Side-by-side comparison of all model performance metrics on the test set. Each panel shows a different metric: ROC-AUC (ensemble methods lead), Recall (Logistic Regression leads at default threshold), Precision (Gradient Boosting and Random Forest lead at default threshold), and F1 score (Logistic Regression leads). No single model dominates on all metrics simultaneously.

Chapter 2

Fairness and Algorithmic Ethics

“The question is not whether your algorithm is biased. It is. The question is whether you know how, and whether you are doing anything about it.”

— Adapted from Barocas, Hardt, and Narayanan (2019)

2.1. Why Fairness Analysis Is Non-Optional

A model that achieves good *average* performance may perform systematically worse for specific demographic groups. In a healthcare context, this means that patients from underserved populations receive worse algorithmic risk stratification, potentially resulting in differential access to post-discharge follow-up resources. This would replicate and amplify existing healthcare inequities rather than addressing them.

The landmark study by Obermeyer, Powers, Vogeli, and Mullainathan (2019) demonstrated that a widely used commercial healthcare algorithm assigned systematically lower risk scores to Black patients than to equally ill White patients, because the algorithm used healthcare spending as a proxy for health need and Black patients historically received less care at equal levels of illness, meaning the same illness level was reflected in lower spending in the training data.¹ The bias was not introduced by the algorithm; it was embedded in the data and then amplified by deployment.

In this chapter we conduct a structured fairness audit of the Logistic Regression model (which is most clinically useful due to its higher recall, and is used across all subgroup analyses for consistency) across three demographic dimensions: race, age group, and gender.

¹The Obermeyer et al. (2019) paper is available at <https://science.sciencemag.org/content/366/6464/447>. It is one of the most important empirical papers on algorithmic bias in healthcare.

2.2. Subgroup Performance by Race

Table 2.1. Logistic Regression performance on the test set stratified by race. Sample sizes reflect the racial composition of the cohort (Caucasian: 74%, African American: 20%, all others <3%). Performance disparities are interpreted in the context of subgroup sample sizes and prevalence rates.

Race	N	Prev.%	F1	Recall	Precision	ROC-AUC
African American	2,730	7.4	0.225	0.550	0.142	0.711
Asian	41	17.1	0.476	0.714	0.357	0.710
Caucasian	7,053	7.2	0.212	0.528	0.133	0.693
Hispanic	190	7.4	0.200	0.429	0.130	0.663
Other	104	4.8	0.205	0.800	0.118	0.897

Several findings require careful interpretation:

Caucasian and African American patients

The two groups with sufficient sample sizes for reliable comparison are Caucasian ($n = 7,053$) and African American ($n = 2,730$). The model performs marginally *better* for African American patients (ROC-AUC 0.711 vs. 0.693) and achieves higher Recall (0.550 vs. 0.528). At first glance, this might seem reassuring. However, this finding does not mean the model is unbiased for this group, it simply means the model's *ranking* is slightly better. Disparities in absolute predicted probabilities (calibration) are not captured by ROC-AUC.

Furthermore, the 7.4% prevalence in African American patients vs. 7.2% in Caucasian patients suggests similar underlying readmission rates in this dataset. Any substantial difference in model performance between these groups would be attributable to model behaviour rather than disease epidemiology, making it a direct fairness concern.

Hispanic patients

Hispanic patients show the lowest ROC-AUC (0.663) and the lowest Recall (0.429) among all groups with interpretable sample sizes. The model misses 57% of true readmissions in this group compared to 47% in the African American group and 47% in the Caucasian group. This disparity is clinically meaningful: Hispanic patients who will be readmitted are less likely to receive a high-risk flag and therefore less likely to receive enhanced post-discharge support.

Several hypotheses could explain this disparity: differential documentation practices across hospital systems serving Hispanic populations, differences in the distribution of ICD-9 codes, or genuinely different patterns of readmission risk that the model does not capture due to underrepresentation in the training data.

Asian and Other subgroups

The sample sizes for Asian ($n = 41$) and Other ($n = 104$) are too small to draw reliable conclusions. The extremely high ROC-AUC for the “Other” group (0.897) almost certainly reflects the small-sample instability of AUC estimates when the number of positive examples is in single digits. These results should

not be interpreted clinically.

2.3. Subgroup Performance by Age Group

Table 2.2. Logistic Regression performance stratified by age group. The Elderly (≥ 80) group shows the lowest ROC-AUC (0.648), the lowest F1 (0.196), and a considerably lower Recall (0.458) compared to other age groups.

Age Group	N	Prev.%	F1	Recall	Precision	ROC-AUC
Young (<40)	908	6.5	0.198	0.542	0.121	0.705
Middle (40–60)	3,235	6.6	0.199	0.533	0.122	0.702
Older (60–80)	4,889	7.4	0.234	0.556	0.148	0.707
Elderly (≥ 80)	1,467	8.0	0.196	0.458	0.124	0.648

The most clinically concerning finding is the **performance gap for elderly patients (≥ 80)**. This group has the highest prevalence of readmission (8.0%), reflecting their elevated clinical risk, but the model performs *worst* precisely for this group: ROC-AUC of 0.648 vs. 0.702–0.707 for younger groups.

The likely explanation is that elderly readmission is driven disproportionately by factors outside the feature set: falls, delirium, caregiver capacity, social isolation, complex polypharmacy management, and functional decline. The in-hospital clinical record captures acute severity but not the post-discharge environment that drives readmission for this group. This is a data limitation that no modelling strategy can overcome.

The clinical implication is serious: the group with the highest readmission risk and the greatest potential benefit from targeted interventions is also the group where the model is least reliable. Deploying this model as a primary triage tool for elderly patients would be inadvisable without supplementary clinical review.

2.4. Subgroup Performance by Gender

Table 2.3. Logistic Regression performance stratified by gender. The male–female performance gap is modest but consistent across all metrics.

Gender	N	Prev.%	F1	Recall	Precision	ROC-AUC
Female	5,764	7.4	0.213	0.513	0.135	0.688
Male	4,735	6.9	0.215	0.560	0.133	0.707

The gender gap in ROC-AUC (0.688 vs. 0.707) and Recall (0.513 vs. 0.560) is modest but consistent in direction. Whether this represents genuine biological or clinical differences in how diabetes and

its complications present across genders, or a modelling artefact from differential representation or documentation, requires further investigation beyond the scope of this dataset.

2.5. Sources of Unfairness

A critical distinction in fairness analysis is between *algorithmic bias* (the model introducing disparities not present in the data) and *data-inherited bias* (the data encoding historical inequities that the model faithfully reproduces). Mehrabi, Morstatter, Saxena, Lerman, and Galstyan (2021) provide a comprehensive taxonomy of these sources.

In this project, the available evidence points to data-inherited bias as the dominant explanation for observed disparities:

1. The Hispanic performance gap (lower AUC and Recall) likely reflects differential representation in the training data: with <2% of training examples being Hispanic, the model has seen fewer examples of Hispanic patients' readmission patterns.
2. The elderly performance gap likely reflects a feature-set limitation: the features that predict readmission for elderly patients (social and functional factors) are simply not present in the dataset.
3. The slight male advantage in ROC-AUC may reflect the higher prevalence of specific documented clinical conditions (cardiovascular disease, renal disease) in the training data for male patients.

The model is using sensible clinical features, but those features were collected in a healthcare system with unequal access, leading to differential data quality.

2.6. Feedback Loop Risks

If this model were deployed to allocate post-discharge follow-up resources (phone calls, home visits, pharmacy checks), a dangerous feedback loop could emerge:

1. The model assigns lower risk scores to Hispanic and elderly patients due to its lower performance on these groups.
2. These patients receive less follow-up support.
3. Some of these patients are readmitted for preventable reasons.
4. Future model retraining on data from this deployment environment learns that Hispanic and elderly patients with these characteristics *were* readmitted even without follow-up.
5. The model's bias is reinforced, not corrected.

Mitigation of these risks include prospective fairness monitoring, mandatory human override capabilities for high-risk subgroups, and regular retraining with fairness constraints.

2.7. Disparate Impact Analysis

A common operational definition of disparate impact is the **four-fifths rule**: if the positive prediction rate for a protected group is less than 80% of the rate for the group with the highest prediction rate, disparate impact is indicated.²

Applying this to Recall:

$$\text{Disparate Impact Ratio} = \frac{\text{Recall}_{\text{Hispanic}}}{\text{Recall}_{\text{African American}}} = \frac{0.429}{0.550} = 0.780 \quad (2.1)$$

This ratio of 0.780 falls below the 0.80 threshold, indicating that the model provides meaningfully less recall for Hispanic patients than for the best-performing racial group. Formal disparate impact is indicated and would require mitigation before clinical deployment.

2.8. Mitigation Strategies

Several strategies could reduce the observed fairness gaps:

Rewighted training: Assign higher sample weights to underrepresented or lower-performing groups during training, encouraging the model to minimise errors more evenly across groups.

Post-hoc calibration: Calibrate the model's predicted probabilities separately for each group, ensuring that probability 0.20 means the same readmission risk regardless of group membership.

Fairness-constrained optimisation: Directly incorporate equalised odds or equalised recall constraints into the model's training objective (Hardt, Price, & Srebro, 2016).

Data augmentation: Augment training data for underrepresented groups through targeted data collection, particularly from hospitals serving large Hispanic populations.

None of these strategies can fully compensate for fundamental feature limitations. The long-term solution requires **richer data collection**: social determinants of health, functional status, post-discharge support availability, and cultural and linguistic factors that drive readmission in diverse populations.

²The four-fifths rule originates from the US Equal Employment Opportunity Commission guidelines (1978). Its application to algorithmic systems is discussed in Barocas et al. (2019), Chapter 3.

Chapter 3

Conclusion

3.1. Answers to the Research Questions

We now answer the five research questions posed in at the first chapter:

RQ1: Can the model achieve ROC-AUC ≥ 0.75 ?

No. The best test-set ROC-AUC is 0.6855 (Gradient Boosting, 95% CI: [0.666, 0.705]). This is a consistent, well-estimated result with narrow confidence intervals. The gap to 0.75 is attributable primarily to a feature-set limitation: the patients most likely to be readmitted but not caught by the model are those whose readmission is driven by social and post-discharge factors not present in the clinical record.

RQ2: Do ensemble methods outperform Logistic Regression?

Yes, on ROC-AUC; not at the default threshold. All three ensemble methods achieve higher ROC-AUC than Logistic Regression (0.679–0.686 vs. 0.662). McNemar’s test confirms these differences are statistically significant. However, at the default $\tau = 0.5$ threshold, Logistic Regression achieves substantially higher recall (0.58) than the ensemble methods because its probability calibration places more predictions near the decision boundary.

RQ3: What are the most important predictors?

Prior hospitalisation history. SHAP analysis confirms that `number_inpatient`, `total_prior_visits`, and `had_prior_inpatient` are the dominant predictors. Secondary predictors include `time_in_hospital`, `num_medications`, and `number_diagnoses`. All top predictors are clinically plausible.

RQ4: What threshold maximises clinical utility?

Clinical context determines the threshold. At $\tau = 0.30$, the Gradient Boosting model flags 158 patients with precision 0.304 and recall 0.064. The optimal threshold depends on the relative costs of false alarms and missed readmissions, which are organisational decisions outside the scope of this analysis. At minimum, any deployment should operate at a threshold substantially below 0.50.

RQ5: Are there demographic performance disparities?

Yes, measurable and clinically meaningful. The model achieves ROC-AUC 0.663 for Hispanic patients vs. 0.711 for African American patients — a gap of 0.048, corresponding to a disparate impact ratio on recall of 0.78 (below the 0.80 threshold). Elderly patients (≥ 80) have the worst model performance (ROC-AUC 0.648) despite having the highest readmission prevalence (8.0%), representing the most critical deployment fairness concern.

3.2. Closing Remarks

Hospital readmission prediction is an important and genuinely difficult problem. This project demonstrates that structured clinical data enables meaningful but limited predictive accuracy, and that the gap between achievable performance and clinical utility thresholds is partly fundamental to the information available in these features. The pipeline documented here is not the final word on this problem, it is a carefully constructed starting point, with honest acknowledgement of its limitations, that future work can build upon. The ultimate measure of any readmission prediction model is not its ROC-AUC but its impact on patient outcomes when embedded in a clinical workflow.

References

- Barocas, S., Hardt, M., & Narayanan, A. (2019). *Fairness and machine learning: Limitations and opportunities*. fairmlbook.org. (Available at <https://fairmlbook.org>)
- Hardt, M., Price, E., & Srebro, N. (2016). Equality of opportunity in supervised learning. In *Advances in neural information processing systems* (Vol. 29, pp. 3315–3323).
- Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A survey on bias and fairness in machine learning. *ACM Computing Surveys*, 54(6), 1–35. doi: 10.1145/3457607
- Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464), 447–453. doi: 10.1126/science.aax2342
- Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3), e0118432. doi: 10.1371/journal.pone.0118432

Appendix A

Project Repository Structure

The complete project repository is organised as follows:

```
1 diabetes_readmission_analysis/  
2 |-- README.md                # Project overview and setup instructions  
3 |-- requirements.txt          # Python package dependencies  
4 |-- configs/  
5 |   |-- config.py            # Global constants, paths, ICD-9 mappings  
6 |-- data/  
7 |   |-- raw/                 # Downloaded raw CSV and mapping file  
8 |   |-- processed/           # Cleaned, engineered feature matrix  
9 |   |-- splits/              # Train, validation, test CSVs  
10 |-- models/                  # Serialised trained models (.pkl)  
11 |-- notebooks/  
12 |   |-- 00_project_overview.ipynb  
13 |   |-- 01_data_collection.ipynb  
14 |   |-- 02_data_cleaning.ipynb  
15 |   |-- 03_feature_engineering.ipynb  
16 |   |-- 04_eda_univariate.ipynb  
17 |   |-- 05_eda_bivariate.ipynb  
18 |   |-- 06_leakage_and_splits.ipynb  
19 |   |-- 07_baseline_models.ipynb  
20 |   |-- 08_advanced_models.ipynb  
21 |   |-- 09_model_evaluation.ipynb  
22 |   |-- 10_visualizations.ipynb  
23 |   |-- 11_error_analysis.ipynb  
24 |   |-- 12_fairness_analysis.ipynb  
25 |-- reports/  
26 |   |-- figures/             # All saved plot PNG files  
27 |   |-- tables/              # All saved results CSV/JSON files  
28 |-- src/  
29 |   |-- eval_utils.py         # Metrics computation and reporting  
30 |   |-- plot_utils.py        # Reusable plotting functions  
31 |   |-- feature_utils.py      # ICD-9 grouping and encoding helpers
```

Listing A.1. Repository directory tree.

Appendix B

Complete Feature Dictionary

Table B.1 provides a complete dictionary of all features in the final processed feature matrix used for modelling.

Table B.1. Complete feature dictionary for the processed feature matrix. Encoding column: Ords = ordinal integer, Bin = binary 0/1, Cont = continuous, OHE = one-hot encoded.

Feature Name	Encoding	Description
time_in_hospital	Cont	Length of inpatient stay in days [1–14]
num_lab_procedures	Cont	Number of lab tests performed [1–132]
num_procedures	Cont	Number of procedures performed [0–6]
num_medications	Cont	Number of distinct medications [1–81]
number_outpatient	Cont	Prior outpatient visits [0–42]
number_emergency	Cont	Prior emergency visits [0–54]
number_inpatient	Cont	Prior inpatient visits [0–17]
number_diagnoses	Cont	Distinct diagnoses recorded [1–16]
admission_type_id	Cont	Admission type code [1–8]
discharge_disposition_id	Cont	Discharge disposition code [1–28]
admission_source_id	Cont	Admission source code [1–25]
age_ordinal	Ords	Age decade bracket encoded 0–9
gender_binary	Bin	Female=0, Male=1
total_prior_visits	Cont	Sum of outpatient + emergency + inpatient visits
had_prior_inpatient	Bin	1 if any prior inpatient visit
n_medications_active	Cont	Number of medications prescribed (not “No”)
n_medication_changes	Cont	Number of medications changed (Up or Down)
n_medication_increases	Cont	Number of medications increased (Up only)
A1Cresult_encoded	Ords	A1C result: 0=not tested, 1=Norm, 2=>7, 3=>8

Table B.1 – *continued*

Feature Name	Encoding	Description
A1C_tested	Bin	1 if A1C test was ordered
max_glu_serum_encoded	Ords	Glucose serum: 0=not tested, 1=Norm, 2=>200, 3=>300
glucose_tested	Bin	1 if serum glucose was measured
med_changed	Bin	1 if any medication was changed during encounter
on_diabetes_med	Bin	1 if any diabetes medication was prescribed
primary_diag_is_diabetes	Bin	1 if primary diagnosis is diabetes (ICD-9 250.xx)
metformin	Ords	Metformin change: 0=No, 1=Steady, 2=Down, 3=Up
insulin	Ords	Insulin change: 0=No, 1=Steady, 2=Down, 3=Up
glipizide	Ords	Glipizide change (same encoding)
glyburide	Ords	Glyburide change (same encoding)
pioglitazone	Ords	Pioglitazone change (same encoding)
rosiglitazone	Ords	Rosiglitazone change (same encoding)
(+18 more drugs)	Ords	All remaining 18 medication columns
diag_1_cat_*	OHE	Primary diagnosis category (9 binary columns)
diag_2_cat_*	OHE	Secondary diagnosis category (9 binary columns)
diag_3_cat_*	OHE	Tertiary diagnosis category (9 binary columns)
race_*	OHE	Race (4 binary columns after drop-first)

Appendix C

Complete Numerical Results

A.1 Final Test-Set Results (All Models)

Model	Acc.	B.Acc.	Prec.	Rec.	F1	AUC	AP	TP/FN
Dummy	0.9284	0.500	0.000	0.000	0.000	0.500	0.072	0/752
Log. Reg.	0.6497	0.617	0.115	0.580	0.192	0.662	0.140	436/316
Rand. For.	0.8852	0.549	0.171	0.157	0.164	0.679	0.139	118/634
Grad. Boost	0.9282	0.501	0.250	0.001	0.003	0.686	0.151	1/751
XGBoost	0.9284	0.500	0.000	0.000	0.000	0.684	0.151	0/752

A.2 Bootstrap 95% Confidence Intervals

Model	ROC-AUC	CI Lower	CI Upper	Width
Gradient Boosting	0.6855	0.6662	0.7049	0.0387
XGBoost	0.6836	0.6649	0.7024	0.0376
Random Forest	0.6789	0.6603	0.6997	0.0394
Logistic Regression	0.6617	0.6418	0.6813	0.0395

Appendix D

Hyperparameter Specifications

Table D.1. All hyperparameter settings for trained models. Parameters not listed use scikit-learn or XGBoost defaults.

Model	Parameter	Value / Rationale
Logistic Regression	C	1.0 (default L2 regularisation)
	solver	lbfgs
	max_iter	2000
	class_weight	balanced
Random Forest	n_estimators	300
	max_depth	10
	min_samples_leaf	20
	max_features	sqrt
	class_weight	balanced_subsample
	n_jobs	-1 (all cores)
Gradient Boosting	n_estimators	300
	max_depth	4
	learning_rate	0.05
	subsample	0.8
	min_samples_leaf	20
XGBoost	n_estimators	300
	max_depth	4
	learning_rate	0.05
	subsample	0.8
	colsample_bytree	0.8
	scale_pos_weight	≈ 7.9

Appendix E

Reproducibility Guide

Environment Setup

```
1 # Python >= 3.9 required
2 python -m venv venv
3 source venv/bin/activate          # Windows: venv\Scripts\activate
4 pip install -r requirements.txt
```

Listing E.1. Creating a reproducible Python environment.

Key package versions: `scikit-learn>=1.2`, `xgboost>=1.7`, `imbalanced-learn>=0.10`, `shap>=0.41`, `pandas>=1.5`, `numpy>=1.23`, `matplotlib>=3.6`, `seaborn>=0.12`.

Notebook Execution Order

Run notebooks sequentially from 00 to 12. Each notebook reads its inputs from the `data/` subdirectory and writes its outputs to `data/`, `models/`, or `reports/`. Do not skip notebooks; each one depends on outputs from the previous.

All random operations use `RANDOM_SEED = 42` defined in `configs/config.py`. Changing this seed will produce different train-val-test splits and slightly different model results.